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Seat No: ________

No. of Printed Pages: 02

SARDAR PATEL UNIVERSITY
B.C.A. - (3rd SEM - NEP) EXAMINATION- 2026
US03MABCA01: FUNDAMENTALS OF DATA STRUCTURES

Date: 10-2-2026, Tuesday

Time: 02:00 PM to 04:30 PM

NOTE: Right hand figure indicates the marks of each question.

Total Marks: [50]

Q. 1 Multiple Choice Questions (One marks each) **[10]**

1 ________ are the building blocks of program.

(A) Information (B) Data Structure

(C) Data (D) Data Types

2 In row major order, elements of matrix are stored on a __________ basis.

(A) Row by Column (B) Column by Column

(C) Column by Row (D) Row by Row

3 All elements of array are stored in _________ fashion.

(A) No-linear ordered (B) Non-linear unordered

(C) Linear unordered (D) Linear ordered

4 An operation that is used to insert an element on a stack is known as ______.

(A) Push (B) Pop

(C) Peep (D) Change

5 Whenever an element is inserted in the queue ____________.

(A) FRONT incremented by 1 (B) FRONT decremented by 1

(C) REAR incremented by 1 (D) REAR decremented by 1

6 Prefix notation is also called _______.

(A) Suffix notation (B) Reverse polish

(C) Polish Notation (D) Both A & B

7 Leaf node is known as __________ node.

(A) Terminal (B) Root

(C) Parent (D) Non-terminal

8 The nodes which have the same parent are called __________.

(A) Degree (B) Leaf

(C) Height (D) Siblings

9 Which of the following is the type of Singly linked list?

(A) One way list (B) Two way list

(C) Three way list (D) Four way list

10 A linked list is ________ type of data structure.

(A) Linear (B) Non-Linear

(C) Both (A) & (B) (D) None of these

(P.T.O.)

Q. 2 Define array. Explain 1-D array in detail. [10]

OR

Q. 2 Explain primitive and non-primitive Data Structure and operations on them in detail. [10]

Q. 3 Define Queue. List all and explain any three types of Queue with example. [10]

OR

Q. 3 Define Stack. List applications of stack and explain operation on stack in detail. [10]

Q. 4 Define traversal. List all traversal and discuss any one with algorithm. [10]

OR

Q. 4 Explain the representation of binary tree in detail. [10]

Q. 5 Define Linked List. List and explain types of linked lists with example. [10]

OR

Q. 5 What is sorting? List sorting techniques and explain any one of them with algorithm & example. [10]

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